Appendix B. Generation and Optimization of physical adversarial attacks

Appendix B.1. Formalization of Generation and Optimization

(1)Adversarial perturbation optimization

The perturbation δ is typically obtained by solving the following optimization problem, as shown in Eq. (B.1).

$$\delta^* = \arg \min_{\|\delta\|_p \leq \varepsilon} \mathcal{L}\big(f(\theta, x + \delta), y\big) \tag{B.1}$$

where $\mathcal{L}$ is a loss function, ε is the perturbation budget upper bound, and δ satisfies $\|\delta\|_p \leq \varepsilon$ (typically with $p = 2$ or ∞) [47]to ensure that x' is visually similar to x. This constraint ensures that the perturbation is imperceptible to human observers or defense mechanisms, while still being able to cause significant model errors.

(2)Targeted attacks optimization

The objective of targeted attacks can be formalized as follows:

$$\delta^*_{\text{targeted}} = \arg \min_{\|\delta\|_p \leq \varepsilon} \mathcal{L}\big(f(\theta, x + \delta), y_{\text{target}}\big) \tag{B.2}$$

The transformation from targeted to untargeted attacks is achieved by substituting the specific target label y_{target} with the true label y_{true} in the objective function.

(3) Optimization objective of gradient-based attacks

Specifically, the optimization objective of gradient-based attacks can be formulated as:

$$\delta^* = \arg \min_{\|\delta\|_p \leq \varepsilon} \Big(\mathcal{L}_{\text{task}}\big(f(x + \delta), y\big) + \sum_{k=1}^{K} \lambda_k \mathcal{R}_k(\delta, x)\Big) \tag{B.3}$$

where $\mathcal{L}_{\text{task}}$ represents the primary task loss that drives the model to produce incorrect predictions on the adversarial example. $\mathcal{R}_k$ are auxiliary regularization terms, which aim to impose specific attack priors or constraints. $\lambda_k \geq 0$ are hyperparameters.

(4) Optimization objective of substitute model attacks

The optimization objective in substitute model attacks can be formulated as:

$$\delta^* = \arg\min_{\|\delta\|_p \leq \epsilon} \mathcal{L}_\Phi\Big(\mathcal{R}(f_s, x, \delta),\ \mathcal{A}(x; f_s, \mathcal{C})\Big) \tag{B.4}$$

where f_s denotes substitute models, $\mathcal{R}$ denotes the representation extraction function that maps the adversarial example to a specific feature space, $\mathcal{A}$ denotes the anchor generation function that constructs the optimization target based on available context $\mathcal{C}$ (full-information (white/grey-box), zero-label (unsupervised/self-supervised), and black-box settings), and $\mathcal{L}_\Phi$ denotes the loss function that measures the discrepancy between the adversarial representation $\mathcal{R}$ and the generated anchor $\mathcal{A}$.

The loss function $\mathcal{L}_\Phi$ can be formalized using the Kullback-Leibler (KL) divergence[198]:

$$\mathcal{L}_\Phi(\mathcal{R}; \mathcal{A}) = \mathcal{L}_{\mathrm{KL}}(\mathbf{z}, \mathbf{a}) = \sum_{i=1}^{C} \mathbf{z}_i \log\left(\frac{\mathbf{z}_i}{\mathbf{a}_i}\right) \tag{B.5}$$

where $\mathbf{z} = \mathcal{R}_{\text{ensemble}}(x, \delta)$ denotes the averaged probability distribution of the adversarial input across the ensemble, and $\mathbf{a} = \mathcal{A}_{\text{ensemble}}(x)$ denotesthe anchor distribution. This setup encourages the adversarial perturbation δ to maximize the KL divergence between the ensemble's predictions on the adversarial example and the anchor, thereby increasing it s transferability to unknown target models by leveraging the consensus of multiple substitute models.